Data Sciences – CentraleSupélec
Advanced Machine Learning
Course V – Nonnegative Matrix Factorization

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From data to a latent representation

Data matrix: n nonnegative observations with p features,

$$X = [x_1, \ldots, x_n] \in \mathbb{R}_+^{p \times n}.$$

For a latent dimension $r < \min(p, n)$, seek

$$\boxed{X \approx WH}, \qquad W \in \mathbb{R}_+^{p \times r}, \quad H \in \mathbb{R}_+^{r \times n}.$$

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$$x_j \approx W h_j = \sum_{k=1}^{r} w_k H_{kj}.$$

- $w_k = W(:, k)$: a component shared by all observations.
- $h_j = H(:, j)$: the latent representation of observation j.

Course I: learn a representation h_j together with a linear reconstruction map W.

What nonnegativity changes

Unconstrained low-rank approximation:

$$\min_{\text{rank}(Y)\leq r} \frac{1}{2}\|X - Y\|_F^2 \quad \Longrightarrow \quad Y = X_r \text{ (truncated SVD).}$$

PCA uses this principle on centered data.

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Nonnegative matrix factorization:

$$\min_{W\ge 0,\, H\ge 0} \frac{1}{2}\|X - WH\|_F^2.$$

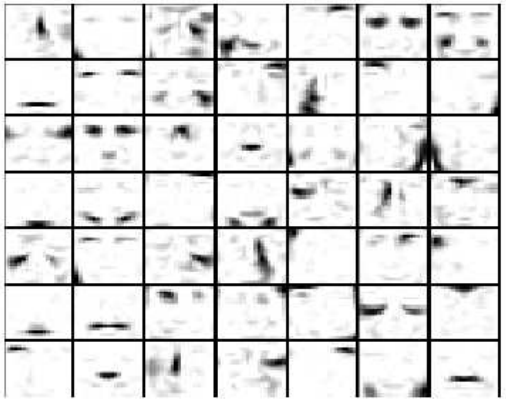
- $\operatorname{rank}(WH) \le r$, with additional constraints on the factors.
- Nonnegative coefficients combine components without cancellation.
- Truncating the SVD or clipping its factors does not solve NMF.

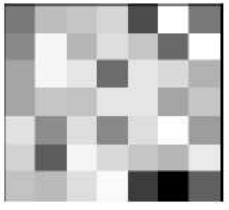
Additive representations can reveal parts, but nonnegativity alone does not guarantee meaningful or unique components.

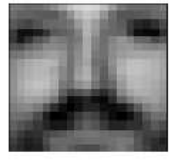
Example: Facial feature extraction

$$\underbrace{X(:,j)}_{j\text{th facial image}} \approx \sum_{k=1}^{r} \underbrace{W(:,k)}_{\text{facial features}} \underbrace{H(k,j)}_{\substack{\text{importance of features} \\ \text{in } j\text{th image}}} = \underbrace{WH(:,j)}_{\substack{\text{approximation} \\ \text{of } j\text{th image}}}$$









CBCL face database [Lee and Seung, 1999]

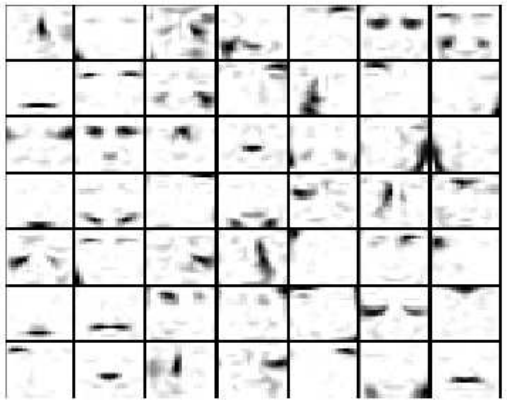
Reading the factors:

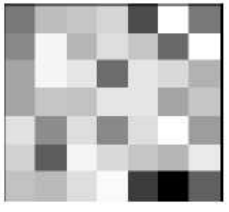
- Columns of W: component images (e.g., parts of a nose or eye).
- H_{kj}: contribution of component k to face j.

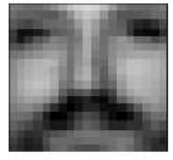
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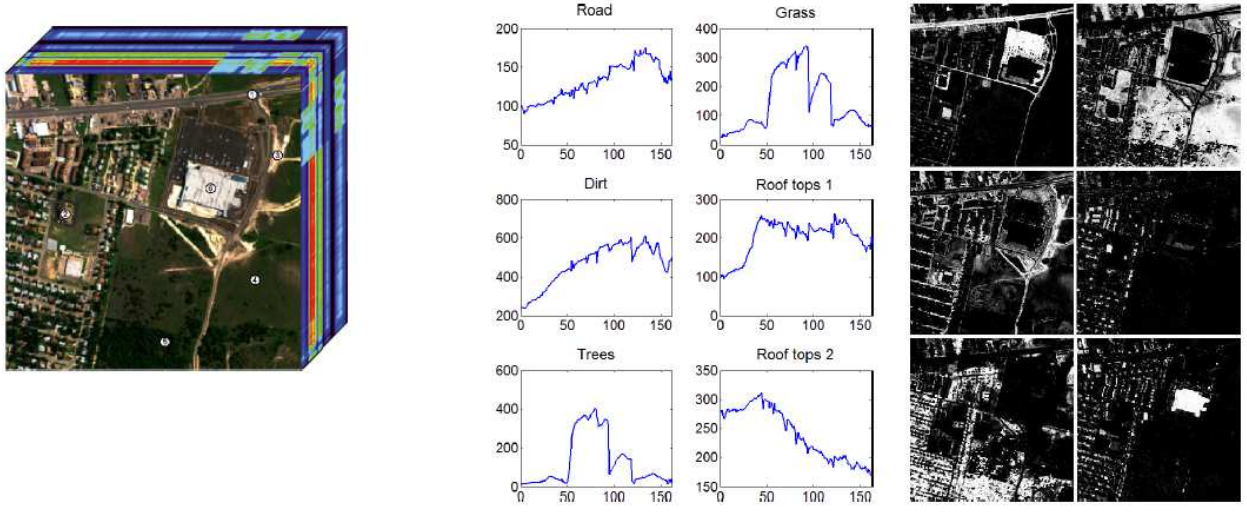
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The lab applies the same model to Olivetti faces. Use the nonnegative pixel intensities, without centering.

Example: Spectral unmixing

$$\underbrace{X(:,j)}_{\text{spectral signature of } j\text{th pixel}} \approx \sum_{k=1}^{r} \underbrace{W(:,k)}_{\text{spectral signature of } k\text{th endmember}} \underbrace{H(k,j)}_{\text{abundance of } k\text{th endmember in } j\text{th pixel}} .$$

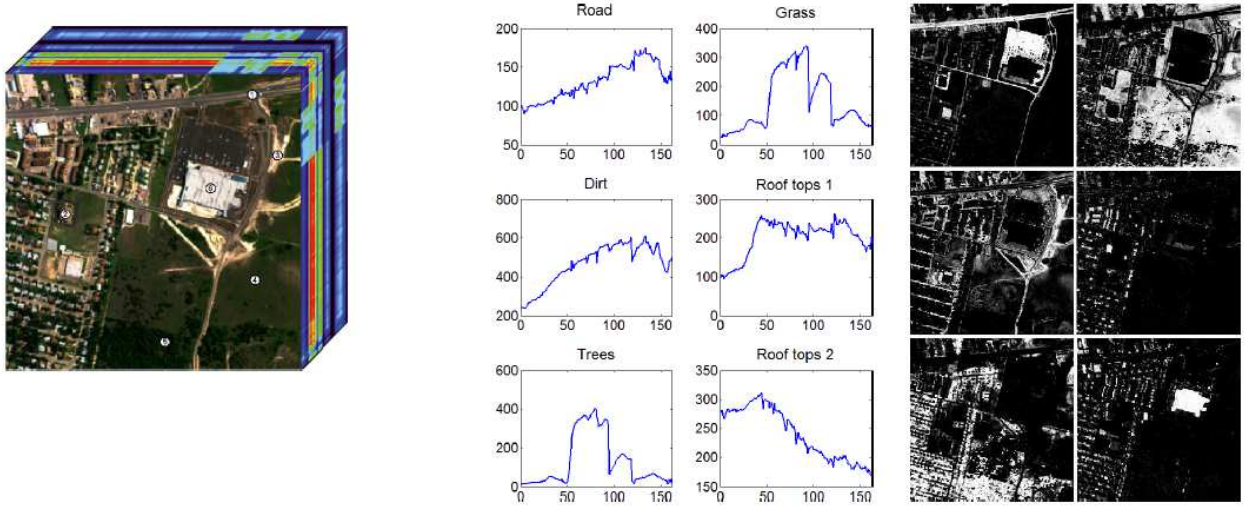


Urban hyperspectral image [Ma et al., 2014]

X: bands $\times$ pixels. Columns of W: endmember spectra.
Rows of H: abundance maps across pixels.

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Rows of H: abundance maps across pixels.

Under a linear mixing model, NMF estimates spectra and coefficients jointly. Abundance fractions additionally require $\sum_k H_{kj} = 1$ and suitable physical assumptions.

Example: Topic modeling in text mining

Term-document matrix: each column is a document and each row a word, with nonnegative counts or TF–IDF weights.

Bag-of-words ignores the ordering of words within a document.

$$\underbrace{X(:,j)}_{j\text{th document}} \approx \sum_{k=1}^{r} \underbrace{W(:,k)}_{k\text{th topic}} \underbrace{H(k,j)}_{\substack{\text{importance of } k\text{th topic} \\ \text{in } j\text{th document}}}$$

Topic decomposition illustration [Blei, 2012]

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Topic decomposition illustration [Blei, 2012]

- Columns of W: word weights defining candidate topics.
- Columns of H: topic weights for each document.

Course IV: a document can combine several topics, whereas a hard clustering assigns it to a single cluster.

Non-uniqueness and normalization

Scaling: for any positive diagonal matrix $D \in \mathbb{R}^{r \times r}$,

$$\boxed{WH = (WD)(D^{-1}H).}$$

Permutation: for any permutation matrix P,

$$WH = (WP)(P^{\top}H).$$

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For each nonzero component, one may normalize

$$s_k = \|w_k\|_1, \qquad w_k \leftarrow w_k/s_k, \quad H_{k,:} \leftarrow s_k H_{k,:}.$$

This fixes a scale convention and preserves the reconstruction. It does not establish identifiability or remove permutation ambiguity.

General rotations need not preserve nonnegativity. Additional ambiguities can still exist beyond scaling and permutation.

From modeling assumptions to a criterion

Euclidean loss:

$$F(W, H) = \frac{1}{2}\|X - WH\|_F^2 = \frac{1}{2}\sum_{i,j}(X_{ij} - [WH]_{ij})^2.$$

Least squares corresponds to an additive i.i.d. Gaussian noise model.

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Generalized KL divergence:

$$\mathcal{KL}(X,Y) = \sum_{i,j}\left(X_{ij}\log\frac{X_{ij}}{Y_{ij}} - X_{ij} + Y_{ij}\right).$$

For independent counts $X_{ij} \sim \text{Poisson}(Y_{ij})$, minimizing KL in Y is maximum likelihood.

$0\log(0/y) = 0$; the scalar divergence is 0 at $(0,0)$ and $+\infty$ for $x > 0, y = 0$.

Course I: model $\rightarrow$ criterion $\rightarrow$ algorithm. KL also serves as a divergence for nonnegative noninteger data.

A nonconvex problem with convex blocks

$$\min_{W \geq 0,\, H \geq 0} F(W,H), \qquad F(W,H) = \tfrac{1}{2}\|X - WH\|_F^2.$$

Fixed W: independent nonnegative least-squares problems,

$$\min_{H \geq 0} F(W,H) = \min_{h_1,\ldots,h_n \geq 0} \sum_{j=1}^{n} \tfrac{1}{2}\|x_j - Wh_j\|_2^2.$$

Fixed H: convex quadratic optimization in W as well.

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Joint nonconvexity: already for $f(w,h) = \frac{1}{2}(1-wh)^2$.
Both $(w,h) = (1,2)$ and $(2,1)$ give $f = 1/2$, but their midpoint gives $f = 25/32 > 1/2$.

Course I: a smooth constrained, nonconvex problem with two convex blocks (biconvex). Block convexity does not imply joint convexity.

Alternating optimization

Starting from $W^{(0)}, H^{(0)} \geq 0$, alternate:

$$W^{(t+1)} \in \underset{W \geq 0}{\text{Argmin}}\ F(W, H^{(t)}),$$

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Exact block minimization gives

$$\begin{aligned} F(W^{(t+1)}, H^{(t+1)}) &\leq F(W^{(t+1)}, H^{(t)}) \\ &\leq F(W^{(t)}, H^{(t)}). \end{aligned}$$

Each block can instead take a cheaper, justified descent step.

Multiplicative updates perform block MM steps. They generally do not solve each nonnegative least-squares subproblem exactly.

Gradients and constrained stationarity

For $F(W,H) = \frac{1}{2}\|X - WH\|_F^2$,

$$G_W := \nabla_W F = (WH - X)H^\top \in \mathbb{R}^{p\times r},$$

$$G_H := \nabla_H F = W^\top(WH - X) \in \mathbb{R}^{r\times n}.$$

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First-order (KKT) conditions:

$$W \geq 0, \quad G_W \geq 0, \quad W \circ G_W = 0,$$

$$H \geq 0, \quad G_H \geq 0, \quad H \circ G_H = 0.$$

Here $\circ$ denotes entrywise multiplication.

At a positive entry, the gradient is zero. At a zero entry, the gradient must be nonnegative. These are necessary, not sufficient, conditions for a joint local minimum.

Euclidean multiplicative updates

Initialization: positive factors W, H.

Each iteration performs the following assignments in order:

$$W \leftarrow W \circ \frac{XH^\top}{WHH^\top}$$

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- All quotients are entrywise; matrix products use standard algebra.
- The H-update uses the *new* W.
- Nonnegative numerators and positive denominators preserve nonnegativity.

A ratio above one increases an entry; a ratio below one decreases it. Zero denominators require explicit handling.

Multiplicative scaling as a gradient step

With W fixed, the H-update can be rewritten entrywise as

$$H^+ = H - \frac{H}{W^\top WH} \circ \nabla_H F(W, H).$$

Interpretation: a gradient step with an entry-dependent scale.

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Zero locking: if $H_{kj} = 0$ and the quotient is defined, then $H^+_{kj} = 0$.

This remains true even if $(\nabla_H F)_{kj} < 0$, when increasing that entry would decrease the objective.

Positive initialization avoids starting on a locked face. The majorization argument explains descent where the updates are well defined; the scaling formula alone does not prove convergence.

MM derivation: a separable Jensen bound

Fix $W \geq 0$ and one data column $x = x_j$. At a reference $\bar{h} > 0$, assume $a_i = (W\bar{h})_i > 0$. Set

$$\alpha_{ik} = \frac{W_{ik}\bar{h}_k}{a_i}, \qquad \sum_k \alpha_{ik} = 1, \qquad (Wh)_i = \sum_k \alpha_{ik} \frac{a_i h_k}{\bar{h}_k}.$$

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Convexity of $z \mapsto (x_i - z)^2$ gives

$$\boxed{\frac{1}{2}\|x - Wh\|_2^2 \leq Q(h \mid \bar{h}) := \frac{1}{2}\sum_{i,k} \alpha_{ik} \left(x_i - \frac{a_i h_k}{\bar{h}_k}\right)^2.}$$

- Equality holds at $h = \bar{h}$ (the surrogate touches the loss).
- Q is separable in the coordinates h_k.

Board work: differentiate each scalar quadratic in Q. Explain why the bound touches the loss at $\bar{h}$.

MM derivation: minimizing the surrogate

Differentiating the surrogate from the previous slide yields

$$\frac{\partial Q}{\partial h_k} = -\sum_i W_{ik} x_i + \frac{h_k}{\bar{h}_k} \sum_i W_{ik} a_i.$$

For positive denominators, its nonnegative minimizer is

$$\boxed{h_k^+ = \bar{h}_k \frac{(W^\top x)_k}{(W^\top W \bar{h})_k}.}$$

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With $f(h) = \frac{1}{2}\|x - Wh\|_2^2$,

$$f(h^+) \le Q(h^+ \mid \bar{h}) \le Q(\bar{h} \mid \bar{h}) = f(\bar{h}).$$

Summing over columns gives the H-step. Transposing the problem gives the W-step.

Course II: as in IRLS, minimize a touching upper bound. Here the surrogate separates the coordinates.

KL multiplicative updates

For $\min_{W,H \geq 0} \mathcal{KL}(X, WH)$, the sequential updates read

$$\boxed{W_{ik} \leftarrow W_{ik} \frac{\sum_{\ell=1}^{n} H_{k\ell} X_{i\ell} / [WH]_{i\ell}}{\sum_{\ell=1}^{n} H_{k\ell}}}$$

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Recompute WH after the W-update. These formulas assume positive reconstruction entries and positive denominators.

The ratio $X_{ij}/[WH]_{ij}$ compares observed and reconstructed intensity. A different loss leads to a different correction.

KL surrogate: the same Jensen principle

Reuse $a_i = (W\bar{h})_i > 0$ and $\alpha_{ik} = W_{ik}\bar{h}_k / a_i$. By convexity of $-\log$,

$$-\log(Wh)_i \leq -\sum_k \alpha_{ik} \log\left(\frac{a_i h_k}{\bar{h}_k}\right).$$

Thus, with $C(x) = \sum_i (x_i \log x_i - x_i)$,

$$\mathcal{KL}(x, Wh) \leq Q_{\mathrm{KL}}(h \mid \bar{h}) = C(x) + \sum_i (Wh)_i - \sum_{i,k} x_i \alpha_{ik} \log\left(\frac{a_i h_k}{\bar{h}_k}\right).$$

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The bound touches at $\bar{h}$ and separates in h_k. Minimizing gives

$$h_k^+ = \bar{h}_k \frac{\sum_i W_{ik} x_i / a_i}{\sum_i W_{ik}}.$$

Board work: differentiate the linear and logarithmic terms. Zero-weight terms are omitted; boundary values use limits.

What descent does and does not guarantee

For well-defined exact MU/MM steps:

$$F_{t+1} \leq F_t, \qquad F_t \geq 0 \quad \Longrightarrow \quad F_t \text{ has a limit.}$$

The same argument applies to the KL objective.

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The same argument applies to the KL objective.

- Convergence of objective values does not imply convergence of W, H.
- A limit of vanilla MU is not automatically a KKT point: boundary behavior and zero locking need analysis.
- Stationarity requires additional assumptions or a suitable safeguarded algorithm.
- Even a stationary point need not be a global, or local, minimum.

Initialization matters. Compare objective values, KKT residuals and component stability across runs.

Convergence caveats: Lin, IEEE TNN, 2007.

Projected gradient exploits the same block structure

With W fixed, $\nabla_H F$ is Lipschitz with $L_H = \|W^\top W\|_2$.

$$\boxed{H^+ = \Big[H - \eta_H W^\top (WH - X)\Big]_+, \quad 0 < \eta_H \leq 1/L_H.}$$

Here $[\cdot]_+ = \max(\cdot, 0)$ entrywise. If $L_H = 0$, the block loss is constant.

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The W-step is analogous, with $L_W = \|HH^\top\|_2$.

- Projection is the proximal map of the nonnegative-orthant indicator.
- A zero entry can become positive when its gradient is negative.

For bounded iterates, cyclic block steps with $\eta_B = 1/(L_B + \tau)$, $\tau > 0$, give sufficient descent and stationary accumulation points for this Euclidean problem.

Courses II–III: combine a gradient step with an operation enforcing the constraint.

Other block strategies

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Hierarchical alternating least squares (HALS): update one component at a time. With

$$R_k = X - \sum_{\ell \neq k} w_\ell H_{\ell,:},$$

the exact column update, for $\|H_{k,:}\|_2 > 0$, is

$$w_k \leftarrow \left[\frac{R_k H_{k,:}^\top}{\|H_{k,:}\|_2^2} \right]_+ .$$

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Clipping an unconstrained least-squares solution generally does not solve NNLS: its quadratic geometry is not the Euclidean projection geometry. Such an ALS heuristic has no general descent guarantee.

Weighted Frobenius loss and missing entries

For known nonnegative weights $\Sigma \in \mathbb{R}_+^{p \times n}$, define

$$F_\Sigma(W, H) = \frac{1}{2}\|\Sigma \circ (X - WH)\|_F^2, \qquad A = \Sigma \circ \Sigma.$$

The corresponding sequential multiplicative updates are

$$W \leftarrow W \circ \frac{(A \circ X)H^\top}{(A \circ (WH))H^\top}, \qquad H \leftarrow H \circ \frac{W^\top(A \circ X)}{W^\top(A \circ (WH))}.$$

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Matrix completion: $\Sigma_{ij} = 1$ on observed entries and 0 elsewhere. Then $A = \Sigma$.

Unobserved ratings are excluded from the loss, rather than fitted as zeros. Their reconstruction quality depends on sampling and model assumptions.

For continuous weights, the square in $A = \Sigma^{\circ 2}$ is essential. MU formulas require positive denominators.

KL model with known row and column gains

Consider known row and column gains $a \in \mathbb{R}^p_{++}$ and $b \in \mathbb{R}^n_{++}$:

$$\min_{W,H \geq 0} \mathcal{KL}\big(X, \mathrm{Diag}(a) WH \,\mathrm{Diag}(b)\big).$$

The predicted count mean is $Y_{ij} = a_i [WH]_{ij} b_j$.

$$W_{ik} \leftarrow W_{ik} \frac{\sum_{\ell=1}^n H_{k\ell} X_{i\ell} / (a_i [WH]_{i\ell})}{\sum_{\ell=1}^n b_\ell H_{k\ell}},$$

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$$W_{ik} \leftarrow W_{ik} \frac{\sum_{\ell=1}^n H_{k\ell} X_{i\ell} / (a_i [WH]_{i\ell})}{\sum_{\ell=1}^n b_\ell H_{k\ell}},$$

$$H_{kj} \leftarrow H_{kj} \frac{\sum_{i=1}^p W_{ik} X_{ij} / (b_j [WH]_{ij})}{\sum_{i=1}^p a_i W_{ik}}.$$

These gains scale the predicted mean. They are not a general entrywise missing-data mask. Positive gains can be absorbed into the factors when no further constraints or penalties are imposed.

Regularized NMF: promoting structure

Course II: add penalties encoding desired properties,

$$\min_{W,H\geq 0} \frac{1}{2}\|X - WH\|_F^2 + \frac{\mu}{2}\|H\|_F^2 + \lambda\|H\|_1 + \frac{\nu}{2}\|W\|_F^2,$$

where $\mu, \lambda, \nu \geq 0$ and $\|H\|_1 = \sum_{k,j} |H_{kj}|$.

- ℓ_1 penalty: encourage sparse activation coefficients.
- Squared Frobenius penalties: control factor magnitudes.

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Scaling must be controlled: if only H is penalized, $W \mapsto cW$, $H \mapsto H/c$ leaves the fit unchanged and decreases its penalty as $c \to \infty$.

Penalizing both factors (e.g., $\mu, \nu > 0$), or constraining their scale, prevents this escape.

Penalties can select a scale, but do not generally give unique factors. Permuting the components still preserves this objective.

Regularized updates preserve nonnegativity

A multiplicative scheme (with positive denominators) is

$$\boxed{W \leftarrow W \circ \frac{XH^\top}{WHH^\top + \nu W}}$$

$$\boxed{H \leftarrow H \circ \frac{W^\top X}{W^\top WH + \mu H + \lambda \mathbf{1}_{r \times n}}.}$$

The sparsity term enters the denominator. Subtracting λ from the numerator can create negative entries.

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Alternatively, a block proximal-gradient step is

$$H^+ = \Big[H - \eta\big(W^\top(WH - X) + \mu H\big) - \eta\lambda\Big]_+,$$

with $0 < \eta \le 1/(\|W^\top W\|_2 + \mu)$ when this bound is nonzero.
Thresholding can create exact zeros. Strictly positive MU iterates may only approach them asymptotically.

Choosing and assessing a factorization

Latent dimension r:

- Too small: merge distinct patterns and underfit the data.
- Too large: redundant components and possible fitting of noise.

The best achievable training loss cannot increase with r, since an extra component can be set to zero.

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Practical checks:

- Compare several initializations at the same r and loss.
- Inspect reconstruction error and components after scale normalization.
- Use held-out entries or observations when appropriate to the task.
- For a new $x \geq 0$, fix W and infer $h \geq 0$ with the same loss.

Low reconstruction error alone does not certify semantic meaning, stability or the correct number of components.

Lab 5: matching the matrix conventions

The notebook receives samples in rows:

$$X_{\text{input}} \in \mathbb{R}_+^{n \times p}.$$

Inside `fit_transform`, it sets `X = X.T`. Hence

$$X = X_{\text{input}}^\top \in \mathbb{R}_+^{p \times n}, \quad \texttt{W_} \in \mathbb{R}_+^{p \times r}, \quad \texttt{H_} \in \mathbb{R}_+^{r \times n}.$$

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Reconstruction in the original input convention:

$$X_{\text{predicted}} = (WH)^\top \in \mathbb{R}_+^{n \times p}.$$

- The code updates W first, then H with the updated W.
- It adds $\delta = 10^{-5}$ to denominators and uses a fixed iteration budget.

The denominator safeguard perturbs exact MU. A fixed iteration count is not a certificate of stationarity.

Lab 5: interpreting faces and topics

Faces: the notebook uses $r = 49$ components.

- Reshape each column of W into a 64×64 component image.
- Display the 49 components in a 7×7 gallery.
- For face j, inspect $H(:, j)$ and reconstruct $WH(:, j)$.
- Fit `faces`; the centered array created in the notebook has negative entries and is not used for NMF.

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Text: the 20 Newsgroups matrix has documents in rows.

- Read the largest entries of each column of W as topic words.
- Read $H(:, j)$ as the topic mixture for document j.

Interpret components and reconstruction together, and keep the transpose convention explicit.

Nonnegative matrix factorization: take-home messages

- $X \approx WH$ learns nonnegative components and latent coefficients.
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- The MM proof establishes objective descent under its assumptions. Stationarity and global optimality are separate questions.

Nonnegative matrix factorization: take-home messages

- $X \approx WH$ learns nonnegative components and latent coefficients.
- The constraints support additive interpretation, with unavoidable scaling and permutation ambiguities.
- The objective is jointly nonconvex but has convex blocks. MU, projected gradient and NNLS exploit that structure.
- The MM proof establishes objective descent under its assumptions. Stationarity and global optimality are separate questions.
- Choosing a loss or regularizer changes both the model and the updates.

A useful NMF combines a suitable criterion, well-defined optimization steps and a careful interpretation of the factors.

References and further reading

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